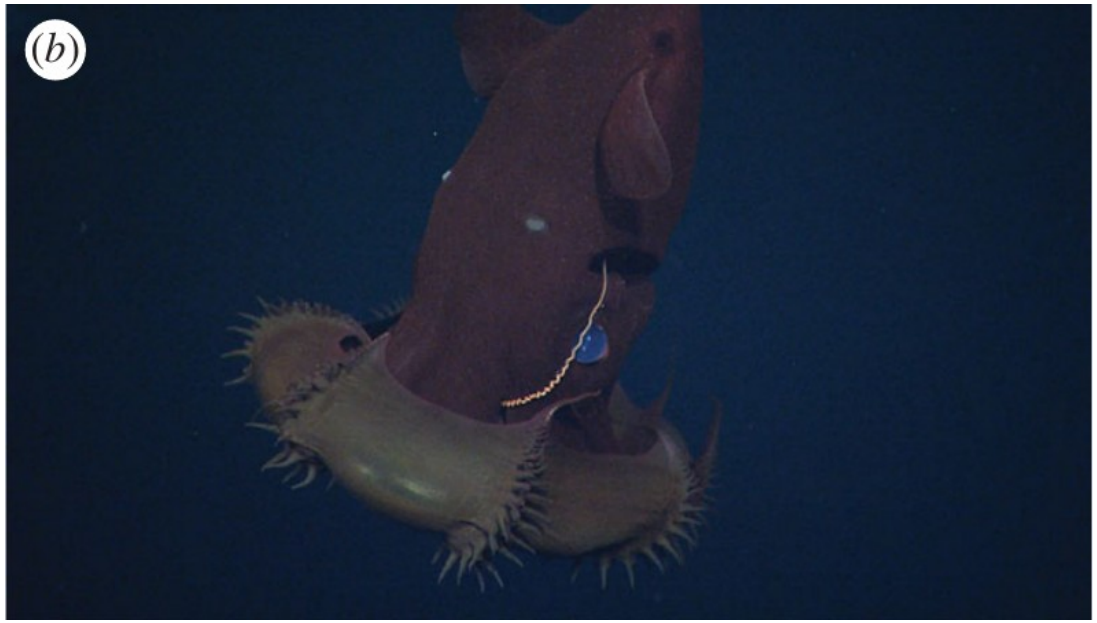
NEUR 411 · NEUROETHOLOGY · FALL 2026

Lecture 100

Deep-sea animals: sensory neuroanatomy, adaptations to darkness, bioluminescent communication, and prey detection in extreme environments

Date TBD

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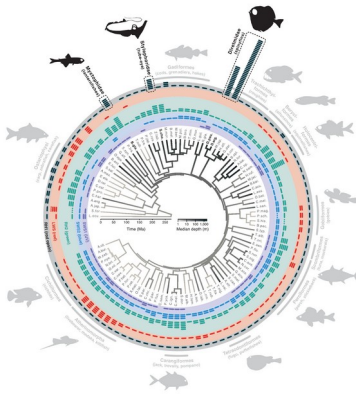
Hoving & Robison (2012), Fig. 1(b). Vampire squid with a retrieved feeding filament.

Darkness changes the available visual signals

At depth, visual information comes from residual sunlight and light produced by organisms. **Bioluminescence** is light generated by a biological chemical reaction. These sources occupy different directions and can carry different information about prey, predators, and nearby animals.

Douglas and colleagues describe a prevailing spectral window near 450–500 nm. Blue light can therefore remain informative even where the surroundings appear dark to a human observer. A visual receptor must absorb the wavelengths that actually reach the animal.

Musilova and colleagues found that some deep-sea fishes possess many rod opsins rather than a single molecular receptor type. Darkness has favored several visual solutions, including molecular diversification, rather than the universal disappearance of useful vision.



Musilova et al. (2019), Fig. 1. Opsin repertoires, expression and spectral tuning.



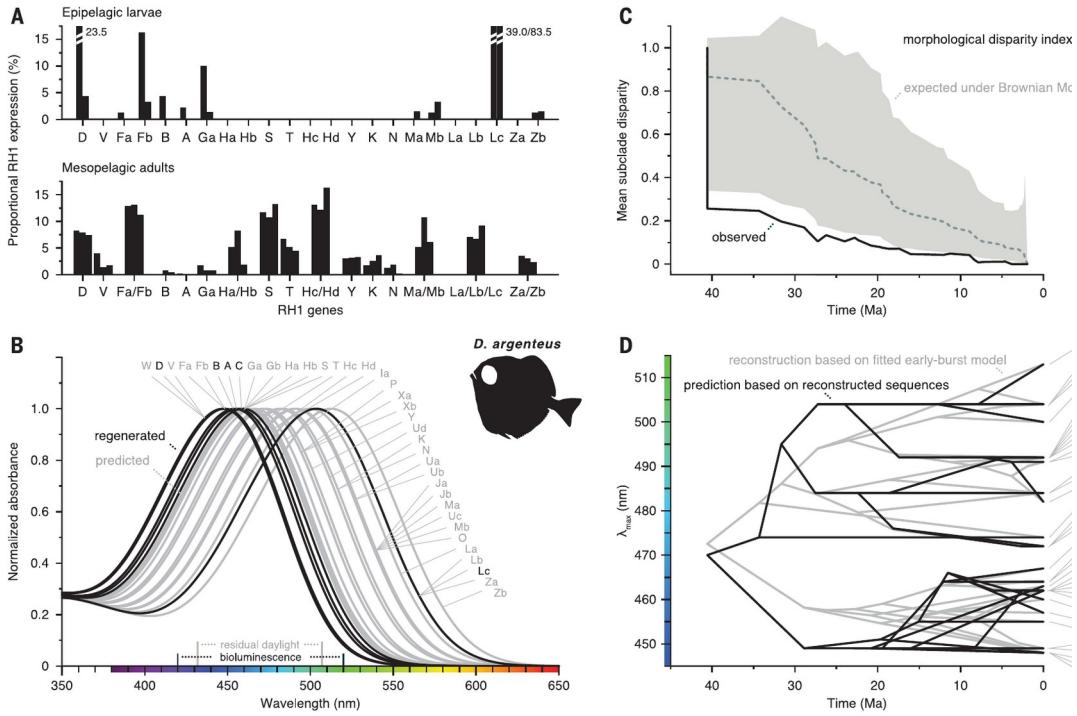
MBARI, natural-history footage. Barreleye (Macropinna microstoma).

An opsin and its chromophore absorb photons

An **opsin** is a light-sensitive receptor protein whose photopigment includes a vitamin A-derived **chromophore**, the molecule that absorbs a photon. Light changes the pigment's molecular configuration and initiates the cellular response called phototransduction.

Musilova and colleagues distinguished rod-associated RH1 opsins from the major cone-associated opsin families. Rods generally support dim-light vision, whereas cones commonly contribute to vision under brighter conditions. These categories connect receptor identity to the conditions under which visual information is acquired.

The study regenerated selected pigments and measured their absorption spectra. This functional preparation connects gene sequence to the wavelength a pigment absorbs most strongly, rather than treating every different gene as an identical sensory receptor.



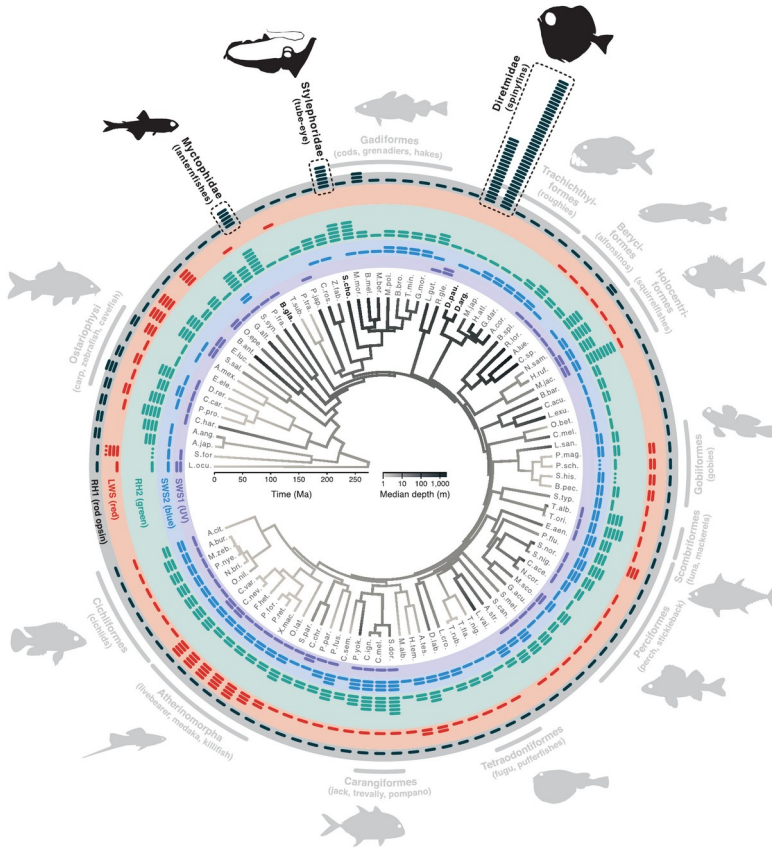
Musilova et al. (2019), Fig. 2. Opsin repertoires, expression and spectral tuning.

Rod opsin expansion evolved in separate lineages

A **gene duplication** creates an additional copy of a gene that can subsequently acquire a different sequence. Musilova and colleagues reconstructed visual opsin repertoires and their evolutionary relationships across fishes, identifying separate expansions of rod-associated RH1 genes.

The expansions occurred in deep-sea lineages including lanternfishes, tube-eyes, and spinyfins. Their separation on the evolutionary tree indicates repeated origins of expanded rod repertoires, rather than inheritance of one expanded repertoire shared by all deep-sea fishes.

The silver spinyfin, *Diretmus argenteus*, possesses 38 RH1 genes and two cone opsins. This unusually large repertoire supplies multiple potential dim-light photopigments, while expression and pigment measurements identify which receptors actually contribute to its sensory equipment.



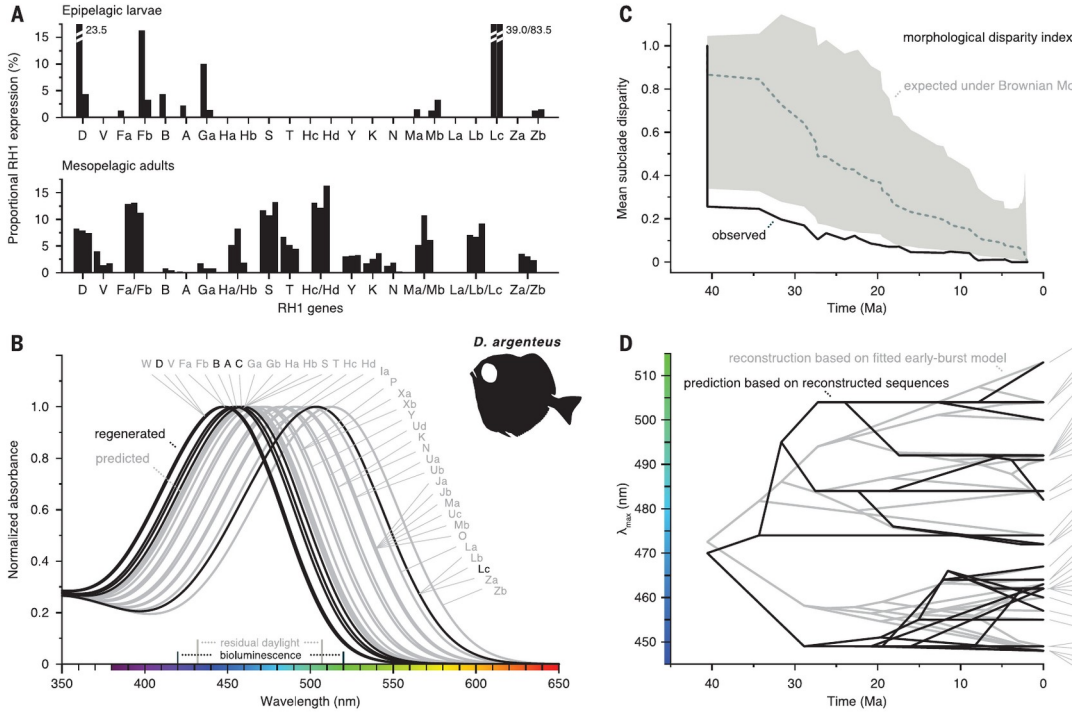
Musilova et al. (2019), Fig. 1. Opsin repertoires, expression and spectral tuning.

Expression changes between larvae and adults

Possessing an opsin gene and expressing it in a retina are different biological events. **Gene expression** is the production of RNA and protein from a gene. Musilova and colleagues compared expressed RH1 repertoires across developmental stages of the silver spinyfin.

Larvae and adults use different combinations of the available rod opsins. The developmental comparison connects changing sensory receptor deployment with life stages that encounter different light environments, rather than assuming that every genomic copy is continuously active.

Spinyfins express up to 14 RH1 genes. Multiple expressed pigments can broaden the range of wavelengths available for detection. Their arrangement and comparison by the nervous system determine how this molecular diversity contributes to visual behavior.



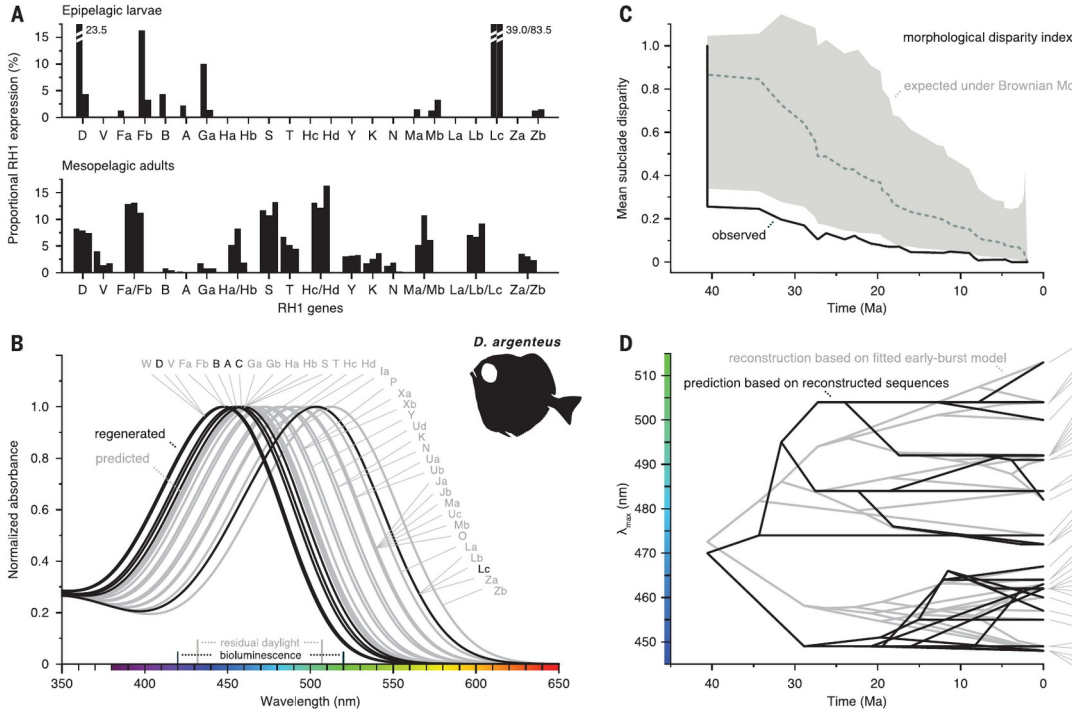
Musilova et al. (2019), Fig. 2. Opsin repertoires, expression and spectral tuning.

Different rod pigments sample different wavelengths

An **absorption spectrum** describes how strongly a pigment absorbs light at each wavelength. Regenerated spinyfin pigments peak at different wavelengths, connecting particular RH1 sequences to differences in the light that can initiate a sensory response.

Measured and reconstructed pigments cover wavelengths associated with residual daylight and deep-sea bioluminescence. The receptor repertoire therefore samples more than one narrow spectral band, potentially helping an animal respond to biologically distinct sources of light.

A peak measured from a regenerated pigment establishes its spectral sensitivity. Color discrimination additionally requires comparison of receptor signals. The hypothesis of rod-based color vision concerns that comparison, while the pigment measurements directly establish the available molecular inputs.



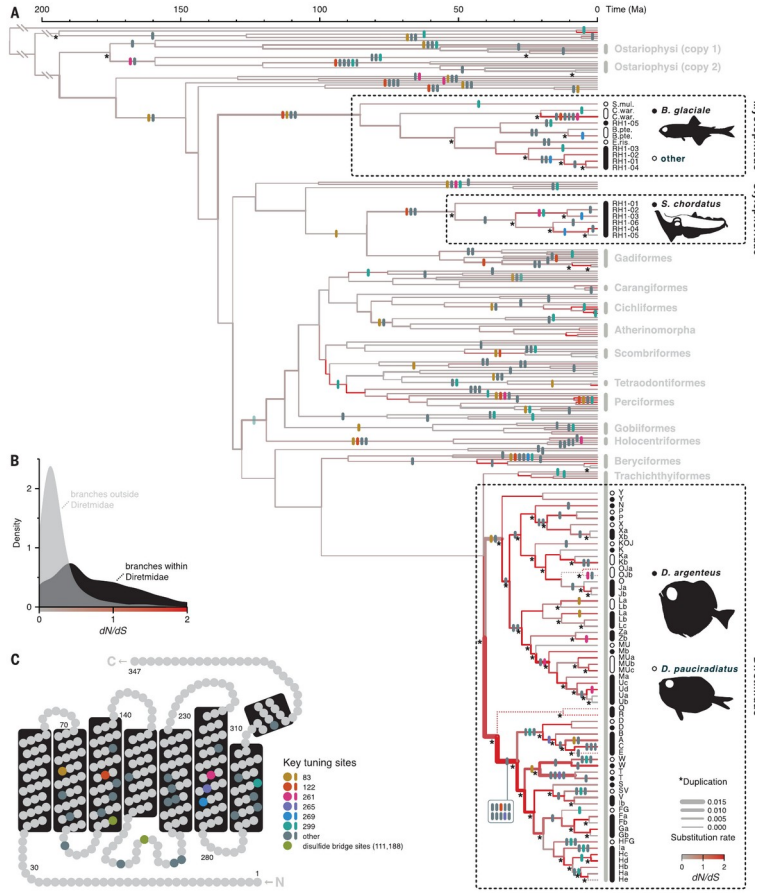
Musilova et al. (2019), Fig. 2. Opsin repertoires, expression and spectral tuning.

Sequence changes alter a receptor’s spectral tuning

Spectral tuning means changing the wavelengths that a photopigment absorbs. Musilova and colleagues mapped amino acid substitutions at positions associated with pigment tuning onto the evolutionary history of RH1 genes in deep-sea fishes.

Duplicated genes diverged in sequence, creating receptors with different absorption properties. Measurements of regenerated pigments provide a functional connection between that sequence variation and the resulting sensory input, rather than relying entirely on predictions from sequence.

Expanded repertoires can therefore change the set of light signals available to a retina. Developmental expression, pigment tuning, and the deployment of those pigments jointly determine how an animal's genomic repertoire becomes a working sensory system.



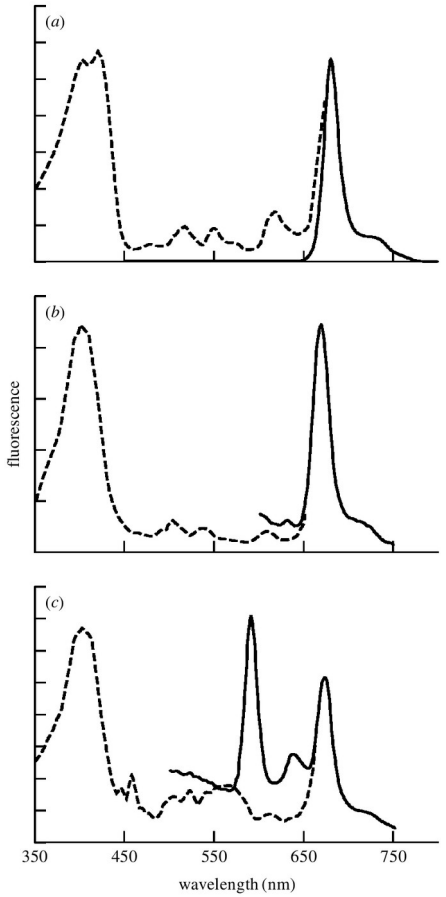
Musilova et al. (2019), Fig. 3. Opsin repertoires, expression and spectral tuning.

Some dragonfishes produce far-red illumination

Three loose-jawed dragonfish genera, *Aristostomias*, *Pachystomias*, and *Malacosteus*, produce far-red light as well as blue bioluminescence. Their **photophores**, organs specialized for light production, include suborbital sources positioned near the eyes.

Far-red emission peaks beyond 700 nm, outside the prevailing blue spectral window. A predator producing this light can obtain useful illumination only if its own sensory system can absorb the emitted wavelengths and generate a corresponding visual response.

Douglas and colleagues compared the mechanisms supporting long-wavelength sensitivity in these fishes. The same ecological problem has different receptor-level solutions: shifted visual pigments in some genera and an additional photosensitizing pigment in *Malacosteus niger*.



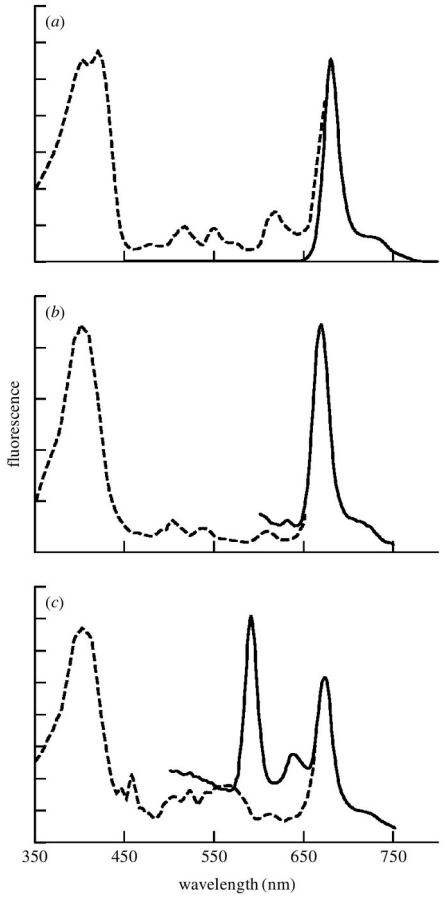
Douglas et al. (2000), Fig. 2. Fluorescence spectra from retinal and dietary material.

A retinal photosensitizer extends light detection

A **photosensitizer** absorbs light and increases the effectiveness of another light-sensitive system. In *Malacosteus niger*, a chlorophyll-derived pigment occurs in the photoreceptor outer segments, the cellular regions that contain visual photopigments.

The dragonfish's own visual pigments peak near 520 and 540 nm. The additional pigment extends useful sensitivity toward far-red wavelengths, supplying a mechanism distinct from simply evolving a conventional visual pigment with a much longer-wavelength absorption peak.

Douglas and colleagues measured fluorescence excitation and emission from retinal preparations. Fluorescence is light emitted after absorption of excitation light. These spectral measurements characterize the additional pigment and connect its molecular properties to the unusual sensory capability.



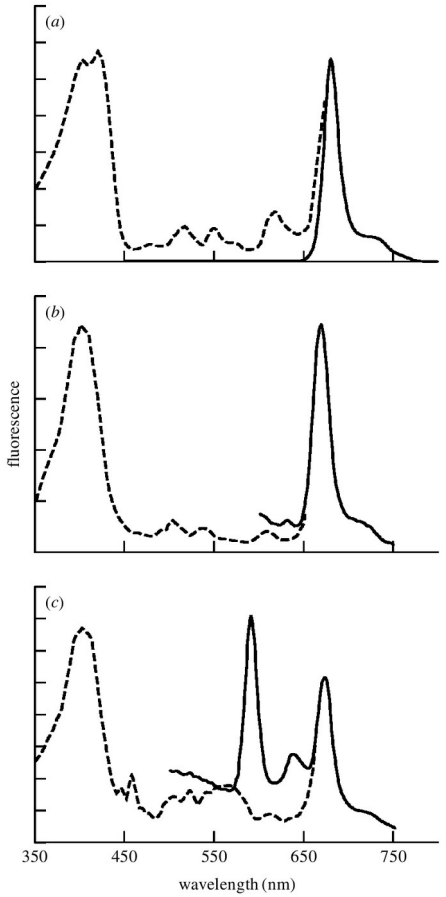
Douglas et al. (2000), Fig. 2. Fluorescence spectra from retinal and dietary material.

Copepod pigments provide a dietary connection

The photosensitizing compound in *Malacosteus* resembles a modified bacterial chlorophyll. Douglas and colleagues compared spectra from the retina, mesopelagic copepods, and material recovered from dragonfish stomachs to investigate how the sensory pigment reaches the animal.

Copepods are crustacean zooplankton and an important component of this dragonfish's diet. Their fluorescence spectra resembled those of the retinal photosensitizer, linking a potential dietary source to a pigment operating within the visual system.

The authors propose acquisition through the food chain rather than manufacture by the fish. This is a dietary-origin hypothesis supported by the spectral similarities. Sensory capability can therefore depend on an ecological supply of material as well as on receptor genes.



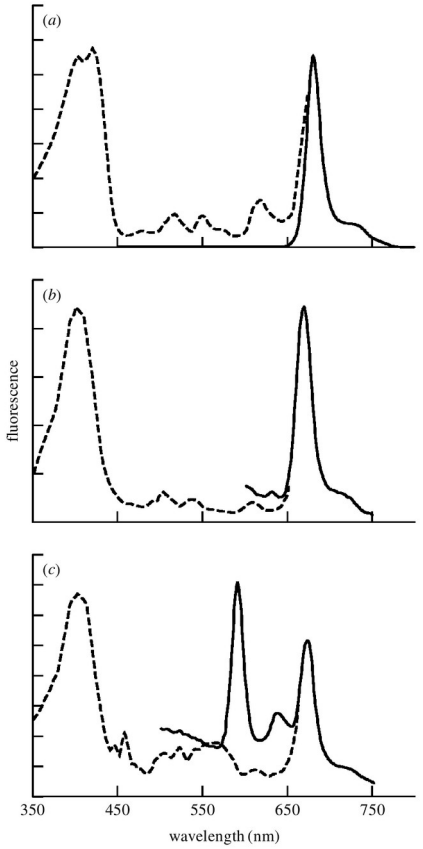
Douglas et al. (2000), Fig. 2. Fluorescence spectra from retinal and dietary material.

Dragonfish genera solve red sensitivity differently

Aristostomias and *Pachystomias* possess several visual pigments shifted toward longer wavelengths, with reported peaks near 515, 550, and 590 nm. Their receptor systems extend sensitivity by combining pigments with different absorption properties.

Malacosteus lacks the longest-wavelength pigment found in those genera and instead uses its retinal photosensitizer. The comparison separates a change in the conventional photopigment repertoire from the recruitment of an additional light-absorbing compound.

Far-red production and sensitivity together create a possible private illumination channel. The hypothesis concerns reduced detection by animals whose receptors are concentrated in the blue range; its mechanism depends on both the emitted spectrum and the receiver's sensitivity.



Douglas et al. (2000), Fig. 2. Fluorescence spectra from retinal and dietary material.



MBARI, natural-history footage. Dragonfish (Idiacanthus antrostomus).

Cockeyed squids have two unequal eyes

The cockeyed squids *Histioteuthis heteropsis* and *Stigmatoteuthis dofleini* possess a large left eye and a smaller right eye. Thomas and colleagues examined their anatomy and orientation using remotely operated vehicle observations in the Monterey Submarine Canyon.

The asymmetry places different visual equipment on the two sides of one animal. Eye size changes the capacity to collect photons, while orientation changes the part of the surrounding light field from which those photons are collected.

The large eye generally faces upward and the small eye downward. These arrangements connect anatomy to two visual tasks: detecting silhouettes against dim downwelling light and detecting bioluminescent sources against the darker water below.



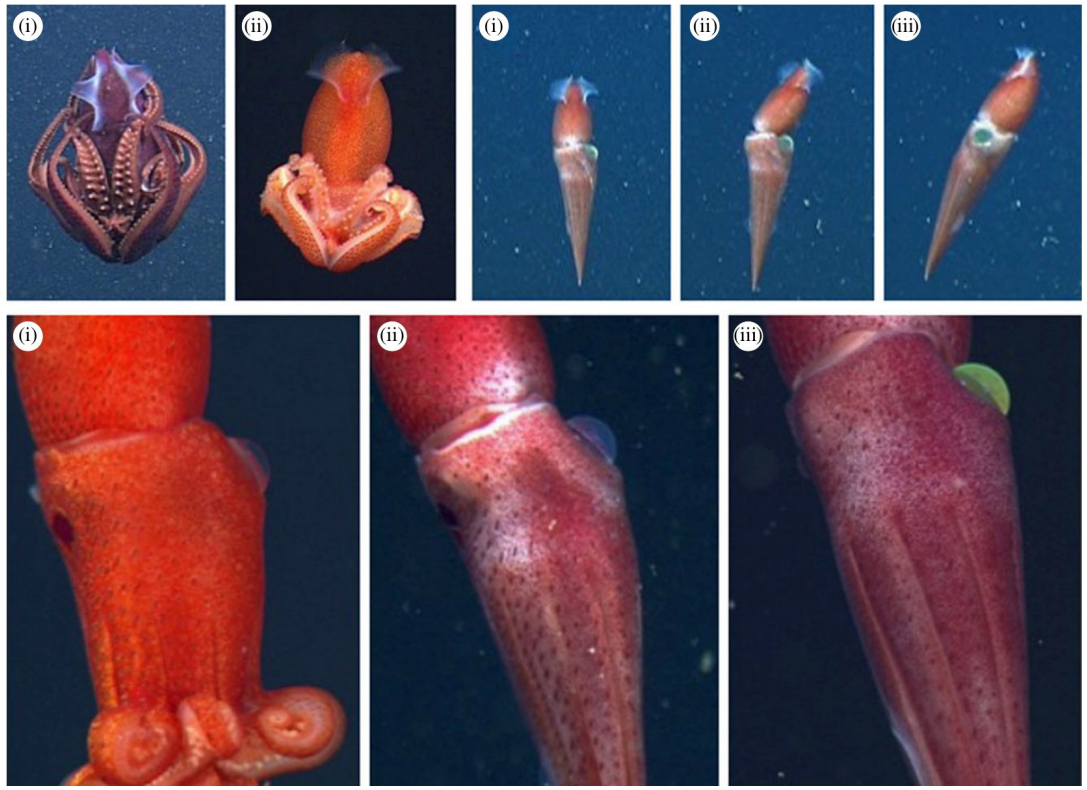
Thomas et al. (2017), Fig. 1. Eye anatomy, orientation and optical specialization.

Body orientation directs eyes toward different scenes

An animal's posture determines which environmental region each eye samples. Thomas and colleagues measured eye and body orientations from in situ footage rather than assuming that preserved anatomy predicts the posture adopted by living squid.

The upward-facing large eye samples the direction of downwelling sunlight. An object passing above can reduce that background illumination, generating a silhouette. The downward-facing small eye samples a region in which a luminous object can provide positive contrast.

This division of visual tasks depends on maintained body orientation. Anatomical asymmetry and posture function together: unequal eyes provide different photon-collecting capacities, and coordinated orientation exposes those eyes to the light conditions suited to their respective tasks.



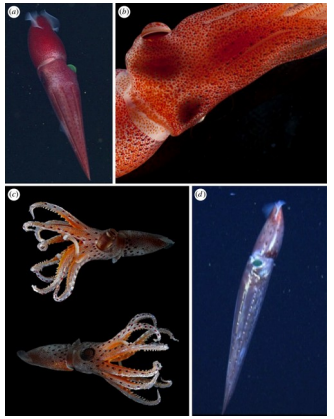
Thomas et al. (2017), Fig. 2. Eye anatomy, orientation and optical specialization.

A pigmented lens filters incoming light

A **lens filter** selectively absorbs part of the light passing toward a retina. Thomas and colleagues recorded yellow pigmentation in the large eye of some cockeyed squids and examined transmission through lenses recovered from captured animals.

Pigmentation changes the spectrum reaching the photoreceptors. The large eye's filtering is associated with the upward-looking visual task, in which the animal encounters residual sunlight and possible bioluminescent signals superimposed on that background.

Filtering can change the relationship between signal and background before receptor activation occurs. The eye's optical structures therefore participate in sensory selection alongside the retina, providing an anatomical mechanism for modifying the visual information delivered to neural tissue.



Thomas et al. (2017), Fig. 1. Eye anatomy, orientation and optical specialization.



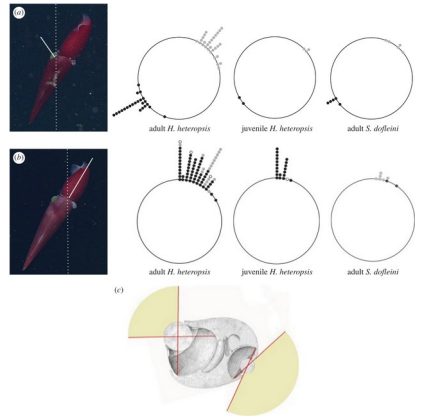
Thomas et al. (2017), Supplementary video 1. Close-up of unequal squid eyes.

Ratcheting turns preserve specialized viewing directions

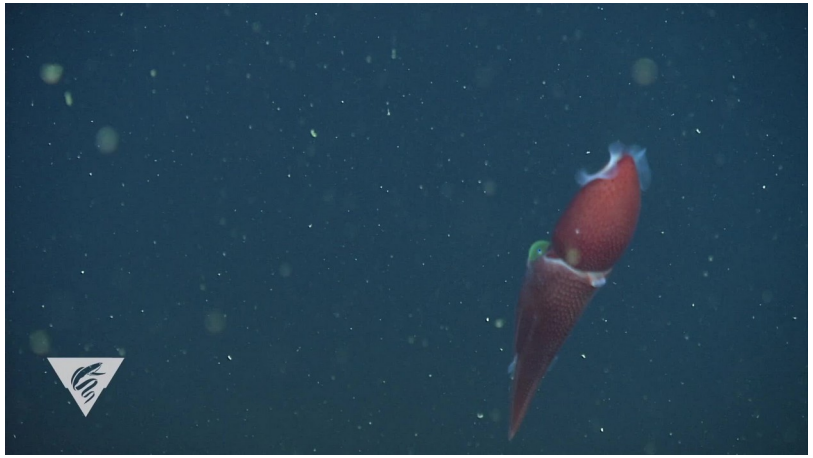
Cockeyed squids make repeated changes in orientation described as **ratcheting turns**. The original recordings document living animals changing body direction while retaining the characteristic arrangement of their unequal eyes relative to the surrounding water column.

A turn changes locomotor orientation and the visual scene at the same time. Maintaining useful viewing directions requires coordination between body posture and the positions of the sensory organs, linking a movement pattern to the acquisition of visual information.

The study compared posture across two histioteuthid species. Their shared eye asymmetry and upward-versus-downward viewing pattern connect a recurring behavioral arrangement to the visual demands imposed by directionally unequal light in the deep ocean.



Thomas et al. (2017), Fig. 3. Eye anatomy, orientation and optical specialization.



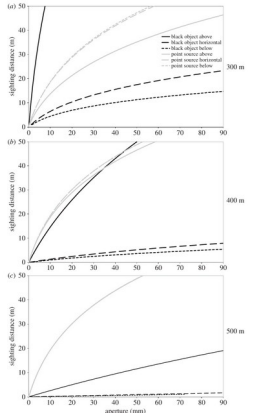
Thomas et al. (2017), Supplemental video 2. Histioteuthis heteropsis makes ratcheting turns.

Large eyes benefit silhouette detection differently

Thomas and colleagues used optical modeling to compare detection of silhouettes with detection of bioluminescent point sources. A **point source** is a luminous object small enough to occupy a restricted angular region in the visual scene.

Increasing eye size improves photon collection, but the benefit depends on the visual task. Their analysis supported a greater benefit of a large upward-looking eye for detecting silhouettes against the dim background than for the downward-looking point-source task.

This task dependence explains why the two eyes need not be equally large. Investing in one enlarged eye can improve the task most sensitive to photon collection while retaining a smaller eye for luminous targets in the darker direction.



Thomas et al. (2017), Fig. 5. Sighting distance depends on aperture, depth and target type.



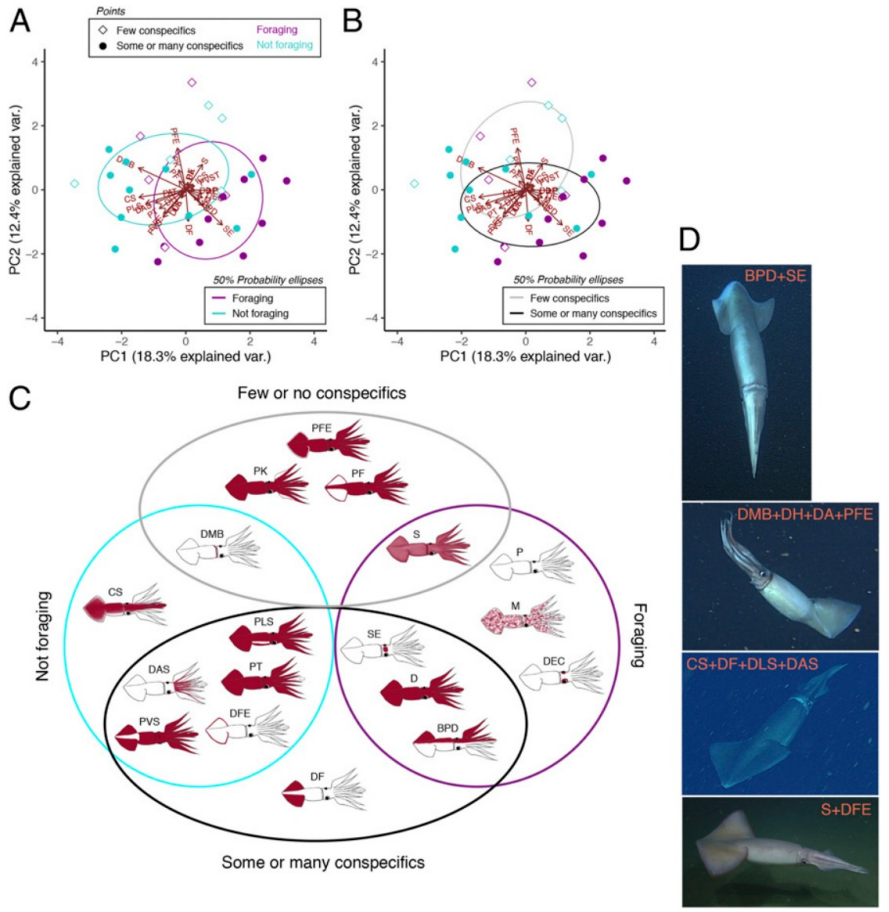
Thomas et al. (2017), Supplemental video 3. Stigmatoteuthis dofleini changes orientation.

Humboldt squid combine body patterns with social context

Humboldt squid, *Dosidicus gigas*, generate conspicuous changes in skin pigmentation during interactions in deep water. Burford and Robison analyzed remotely operated vehicle footage to describe those patterns and relate them to behavior and nearby squid.

A **chromatophore** is a pigment-containing skin structure whose expansion or contraction changes the visible pattern. Combinations of light and dark regions provide a repertoire of signals that can change during an interaction without changing body position.

The analysis related particular components to foraging and social circumstances. Pattern production is therefore treated as organized behavior with context-dependent combinations, rather than as a single undifferentiated flash shared by every activity.



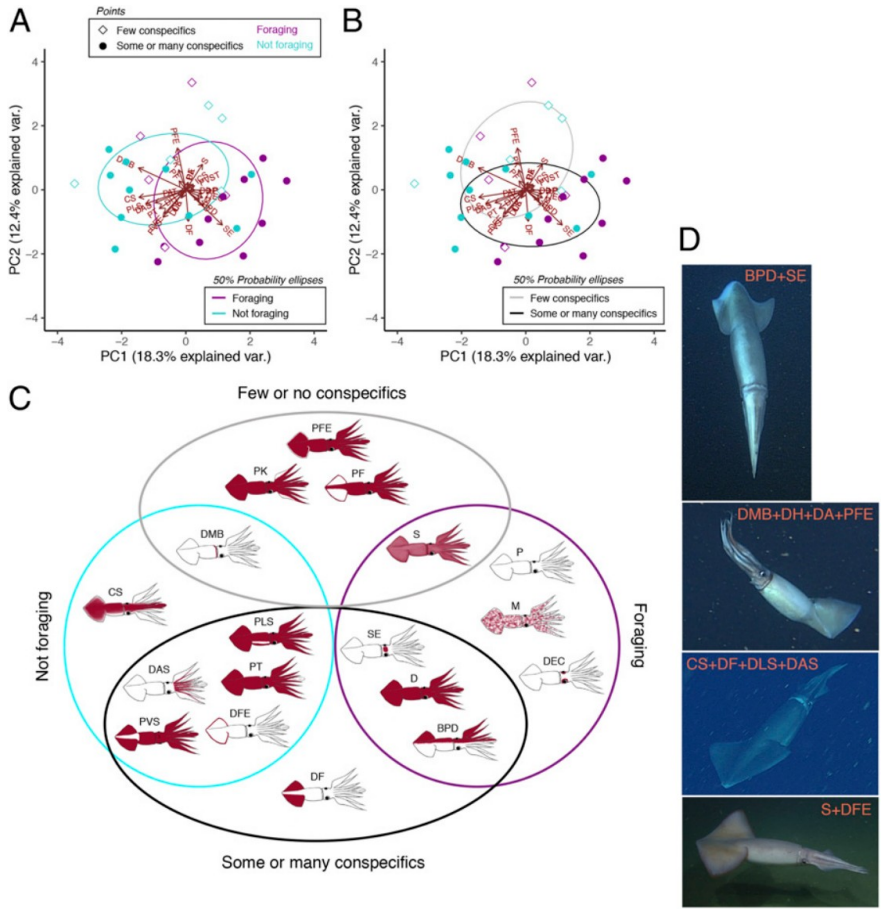
Burford & Robison (2020), Fig. 1. Pigmentation components and behavioral context.

Signal components combine across the body

Burford and Robison described discrete pigmentation components occupying different regions of the Humboldt squid's body. Components can occur together, creating patterns whose organization includes both the identity of a component and its spatial relationship to others.

This **combinatorial signaling** uses combinations of elements to enlarge the available repertoire. Simultaneous patterns across the mantle, head, and arms can distinguish behavioral contexts more effectively than a single whole-body light-versus-dark state.

The researchers scored components from recorded animals and examined their associations with social and foraging conditions. The resulting ethogram, a defined catalog of behavior, provides operational descriptions that connect visible pattern production to repeatable observations of animal activity.



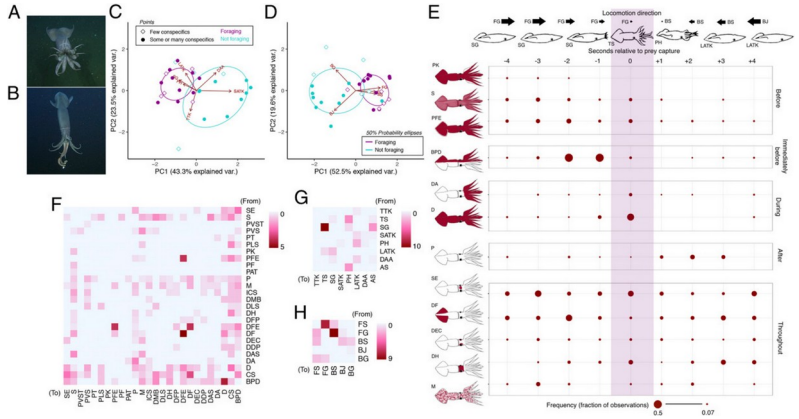
Burford & Robison (2020), Fig. 1. Pigmentation components and behavioral context.

Pigmentation changes accompany prey capture

A foraging squid changes arm and body position while producing pigmentation patterns. Burford and Robison connected particular visual components to the sequence surrounding feeding, using in situ recordings that preserved the animal's natural behavioral context.

Prey capture combines orientation, approach, and coordinated use of the arms. Changes in pigmentation occur within that motor sequence, allowing a nearby animal to receive visual information about the activity of the foraging squid.

The original supplementary movie supplies the recorded movement and pattern transitions. The paper's coded observations connect those transitions to behavioral context, separating the chronology of a feeding event from a static description of the skin pattern.



Burford & Robison (2020), Fig. 2. Pigmentation components and behavioral context.



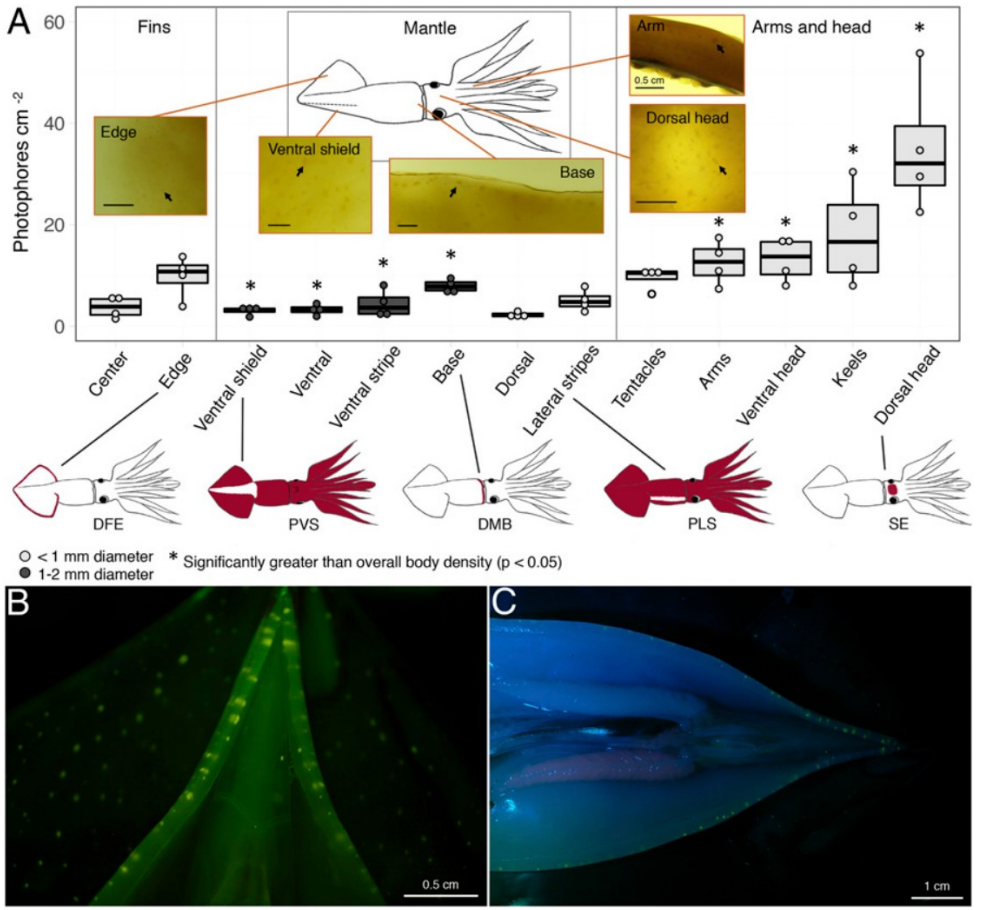
Burford & Robison (2020), Movie S1. Humboldt squid pigmentation during in situ activity.

Photophores can backlight pigmentation patterns

A **photophore** produces light, whereas a chromatophore changes pigmentation. Humboldt squid possess both systems. Burford and Robison proposed that numerous small photophores can provide illumination beneath the changing patterns of the skin.

This **backlighting hypothesis** connects light production to signal visibility in dim water. A dark region can stand out against an illuminated background even when external sunlight supplies little illumination, allowing pigmentation to remain informative at depth.

The anatomical relationship between photophores and pigmentation supplies the proposed mechanism. The footage establishes context-dependent patterning, while the proposed illumination mechanism explains how those patterns could be accessible to receivers under the natural low-light conditions.



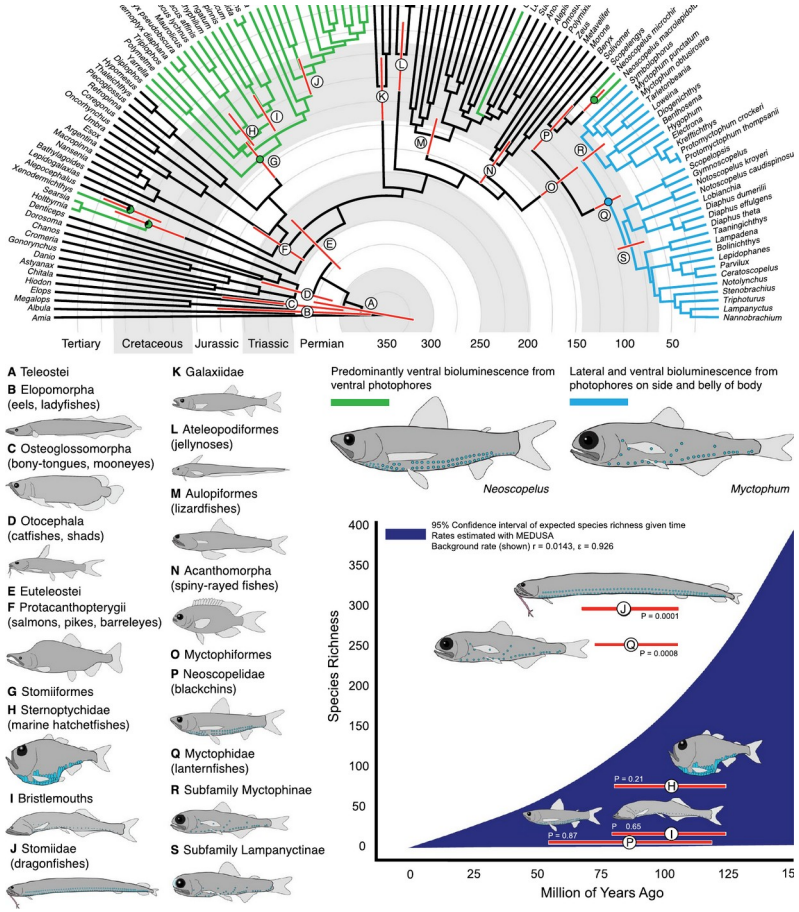
Burford & Robison (2020), Fig. 3. Pigmentation components and behavioral context.

Photophore arrangements differ among fish lineages

Davis and colleagues compared deep-sea fish groups with different uses of bioluminescence. Light can support camouflage, prey attraction, or possible recognition between animals, so the ecological consequences depend on how luminous structures are arranged and used.

Species-specific photophore patterns are arrangements that differ among species. Lanternfishes and dragonfishes include conspicuous examples of patterned light organs, providing potential visual information about the identity of another animal.

A spatial arrangement becomes informative when a receiver can detect and distinguish its features. Comparing these groups connects anatomical organization of luminous organs to hypotheses about communication, while retaining the distinction between recognition signals and light used primarily for other functions.



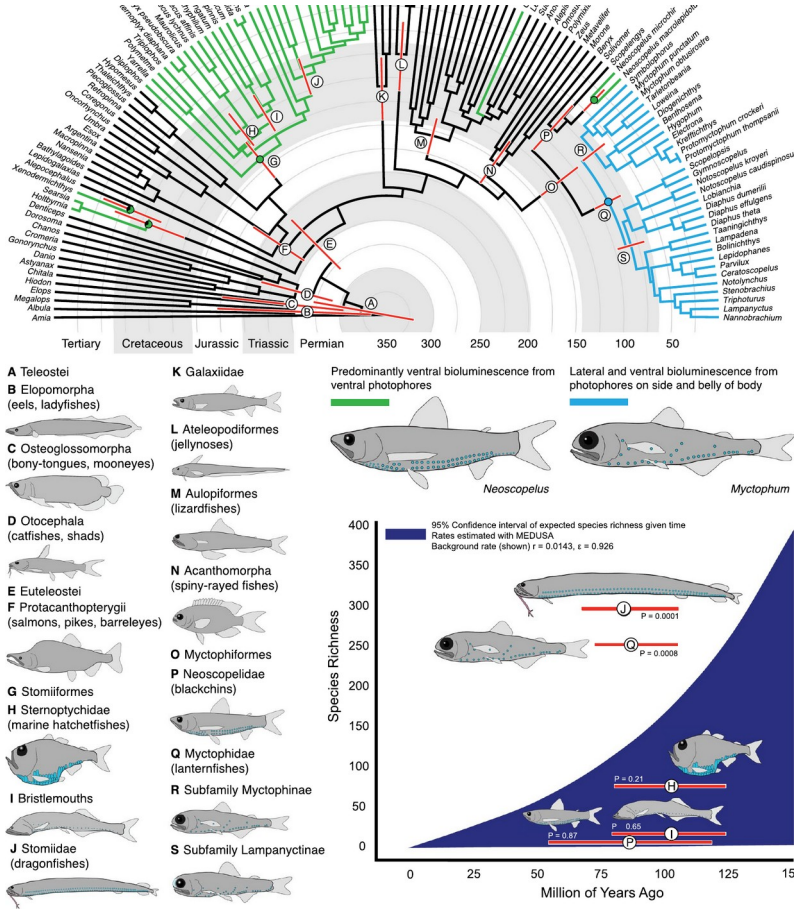
Davis et al. (2014), Fig. 1. Photophore organization and lineage diversification.

Signaling hypotheses connect light to diversification

Diversification is the evolutionary production of distinct lineages through the balance of speciation and extinction. Davis and colleagues compared diversification among deep-sea fish groups differing in the organization and proposed uses of their bioluminescence.

Groups with species-specific luminous structures diversified more rapidly in their comparative analysis than groups using light in arrangements not expected to promote recognition. The association links a sensory signaling trait to patterns of evolutionary diversity.

The proposed mechanism is that recognizable signals can contribute to separation of populations. This is a macroevolutionary hypothesis rather than a direct recording of mate choice; the study connects photophore organization to lineage-level patterns that follow from different communication systems.



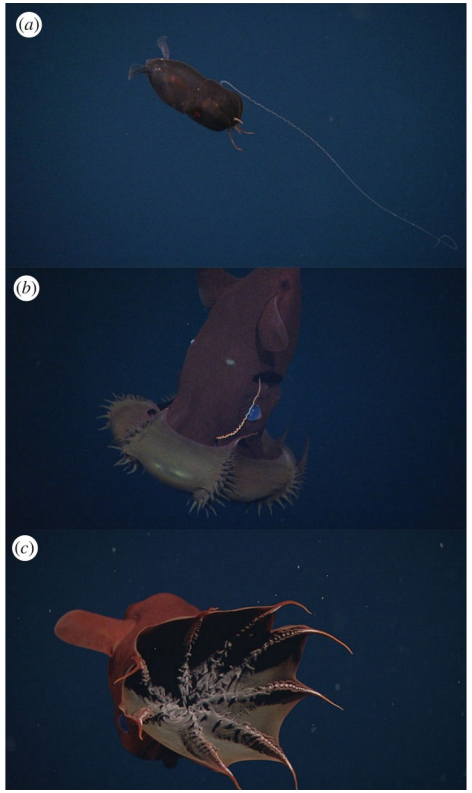
Davis et al. (2014), Fig. 1. Photophore organization and lineage diversification.

Vampire squid collect food with retractile filaments

The vampire squid, *Vampyroteuthis infernalis*, deploys long retractile filaments that differ from its eight arms. Hoving and Robison combined deep-sea observations, feeding trials, digestive contents, and microscopy to examine the function of these structures.

A filament extends into the surrounding water and contacts suspended material. The animal retrieves it through the arms, where material can be gathered into a mucus-associated food mass, connecting collection by a slender appendage to ingestion.

The feeding observations support detritivory, consumption of nonliving organic material, rather than an exclusively predatory interpretation of the animal. A specialized collection structure allows food acquisition from particles and aggregates available in the deep water column.





Hoving & Robison (2012), Fig. 1. Feeding structures and collected material.

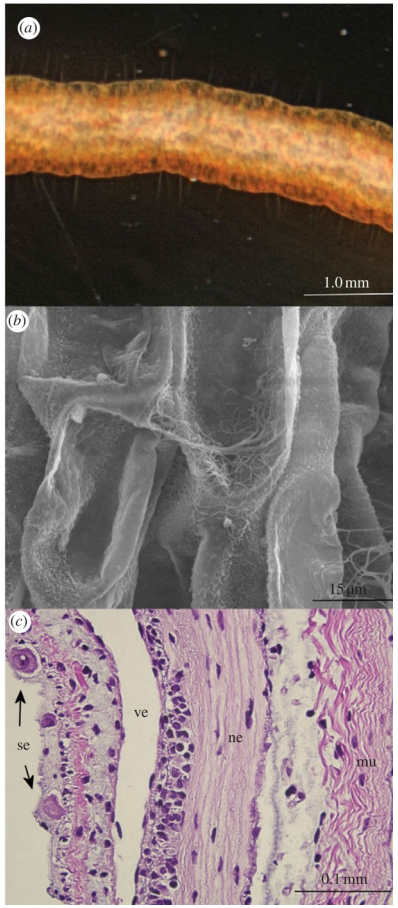
MBARI, natural-history footage. Vampire squid (Vampyroteuthis infernalis).

Filament surfaces combine sensory and collection features

Microscopy revealed distinct cellular regions on vampire-squid filaments. Hoving and Robison described structures interpreted as sensory elements alongside filament surfaces and hairs that contact material suspended in the surrounding water.

Sensory morphology concerns anatomical features capable of receiving environmental information. Their distribution provides a structural basis for detecting contact during collection, linking the filament's interaction with particles to information available to the animal.

Collection also requires retention and transfer of material. Surface features, mucus production, filament retrieval, and arm movements jointly connect contact with suspended food to an ingestible mass, giving the feeding system both sensory and handling components.



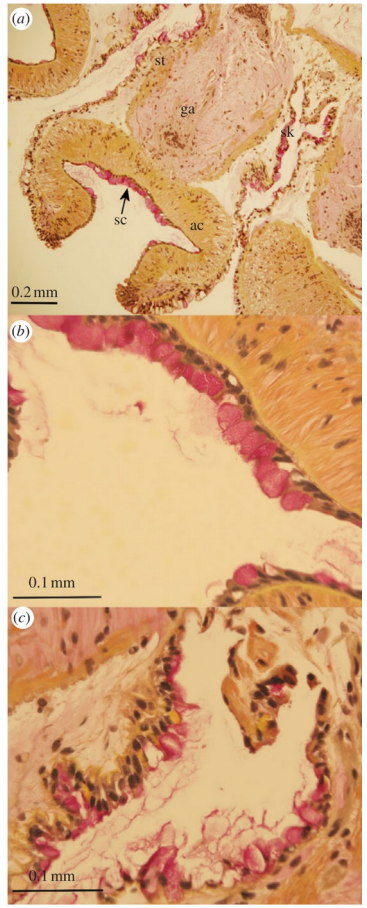
Hoving & Robison (2012), Fig. 3. Filament hairs and sensory-cell morphology.

Mucus and cirri organize collected material

Cirri are finger-like projections along the oral surfaces of the vampire squid's arms. These projections and associated mucus-producing structures differ from the large suckers and capture surfaces characteristic of many actively predatory cephalopods.

Hoving and Robison observed retrieved material handled by the arms and incorporated into mucus-associated masses. Mucus helps retain dispersed particles, converting material collected along a filament into a form that can be moved toward the mouth.

Digestive contents included detrital material and remains associated with the suspended food supply. The agreement between anatomy, handling behavior, and ingested material connects a distinctive cephalopod body plan to a specific mechanism of obtaining food.



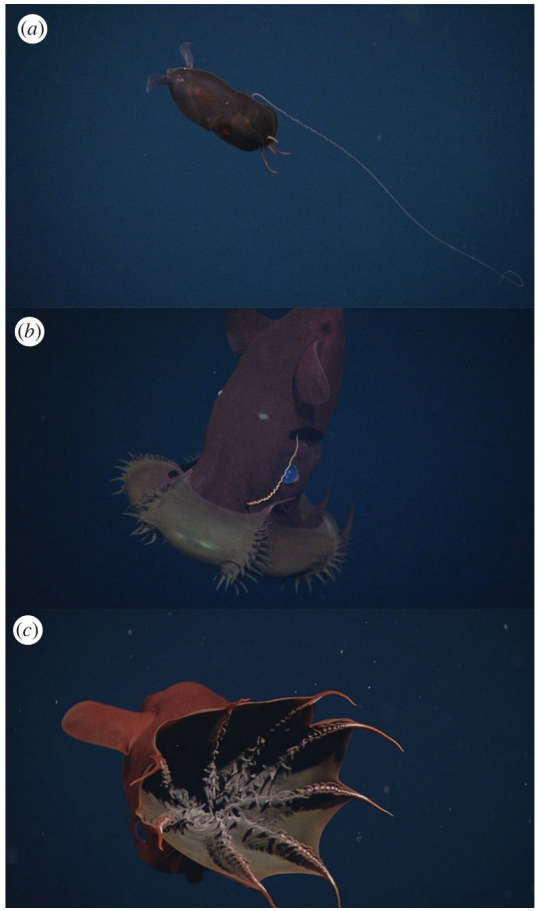
Hoving & Robison (2012), Fig. 2. Arm structures, cirri and mucus-producing cells.

Detrital feeding supports life in oxygen-poor water

An **oxygen minimum zone** is a water layer with unusually low dissolved oxygen. Vampire squid commonly occupy such conditions, where the energetic demands of rapid pursuit and repeated vigorous movement can constrain feeding strategies.

Filament collection allows access to suspended detritus without requiring every food item to be chased and seized. Hoving and Robison documented a feeding mode that connects slow collection movements to the organic material transported through the water column.

The ecological comparison with predatory squid concerns the source and handling of food. Vampire-squid morphology supports an alternative route from environmental particles to ingestion, expanding the range of cephalopod feeding strategies available under deep-water resource and oxygen conditions.



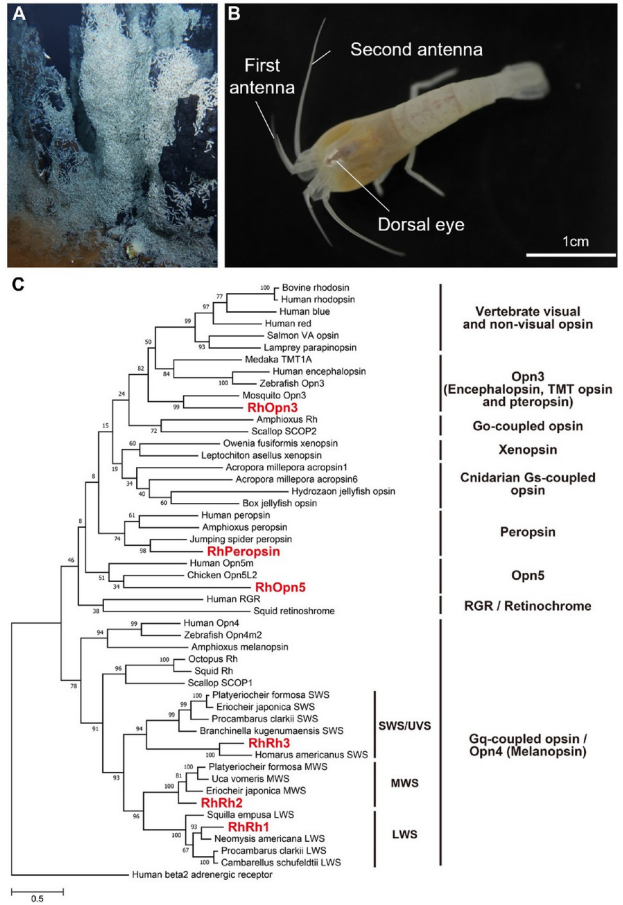
Hoving & Robison (2012), Fig. 1. Feeding structures and collected material.

Vent shrimp retain several light-sensitive receptors

The hydrothermal vent shrimp *Rimicaris hybisae* has an enlarged dorsal eye beneath its carapace rather than ordinary image-forming eyestalk optics. Nagata and colleagues examined this sensory anatomy and identified six opsins, connecting photoreceptor organization to a molecular repertoire operating without sunlight.

Their preparations included purified pigments and cells expressing individual receptors. **Heterologous expression** means producing a receptor in a different cellular system so that its properties can be examined under controlled experimental conditions.

Microvilli are small membrane projections that provide space for photopigments. Microscopy revealed organized microvillar arrays in the dorsal photoreceptors. The receptor repertoire includes visual opsins and Opn5, while experiments with expressed proteins distinguish light reception, cellular signaling, and chromophore regeneration among the six molecules.



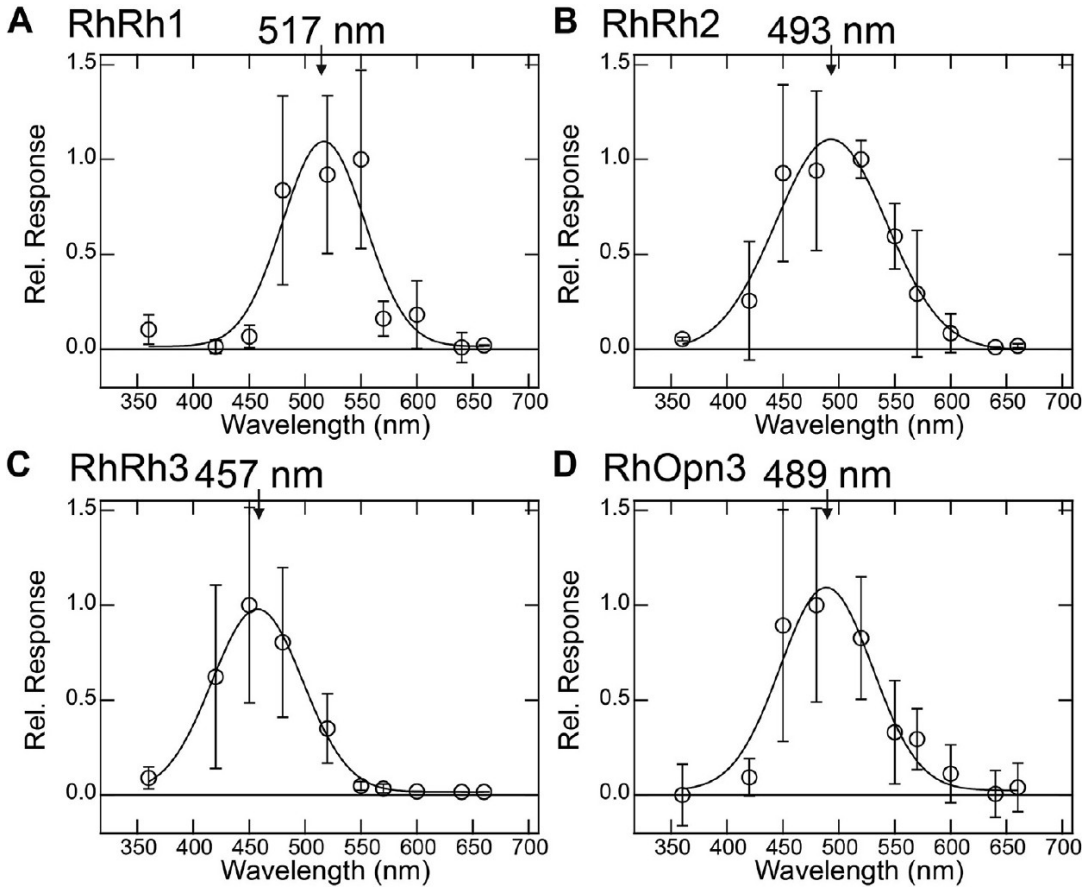
Nagata et al. (2025), Fig. 1. Opsin characterization and light-dependent responses.

Absorption measurements distinguish shrimp pigments

Nagata and colleagues compared spectra of the shrimp's expressed opsins. A spectrum identifies the wavelengths absorbed by a pigment, connecting the receptor's molecular identity to the light that can activate it under the experimental conditions.

The measured pigments respond in the visible range near 457–517 nm. Their different spectral sensitivities connect individual receptor identities to the wavelengths capable of initiating a response, matching molecular light detection to the blue-green signals present around hydrothermal vents.

These measurements supply molecular mechanisms relevant to a sensory system operating without sunlight. They identify available light-sensitive components, while the animal's detection of environmental signals additionally depends on receptor deployment and connections to downstream cellular and behavioral responses.



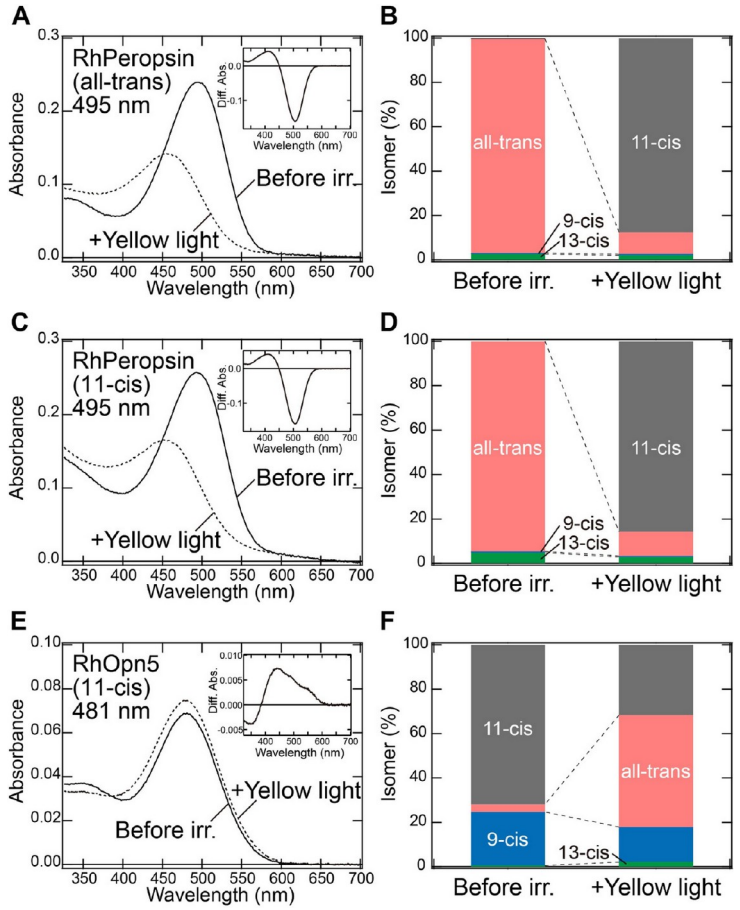
Nagata et al. (2025), Fig. 4. Absorption spectra of four shrimp opsins.

Photochemical states change after illumination

A **photochemical state** is a molecular configuration of a pigment associated with a particular absorption spectrum. Nagata and colleagues exposed shrimp opsins to defined illumination and measured changes in the resulting pigment spectra.

RhPeropsin absorbed most strongly near 495 nm and bound all-trans-retinal. Illumination above 500 nm increased production of 11-cis-retinal. These are different molecular configurations of retinal, and conversion between them supplies a route for producing the chromophore used by other photopigments.

A **photoisomerase** uses light to convert a molecule between configurations. RhPeropsin performed this chromophore-converting function, whereas the other five identified opsins showed light-dependent G-protein activity. The repertoire therefore includes molecular support for pigment regeneration as well as receptors that initiate cellular signaling.



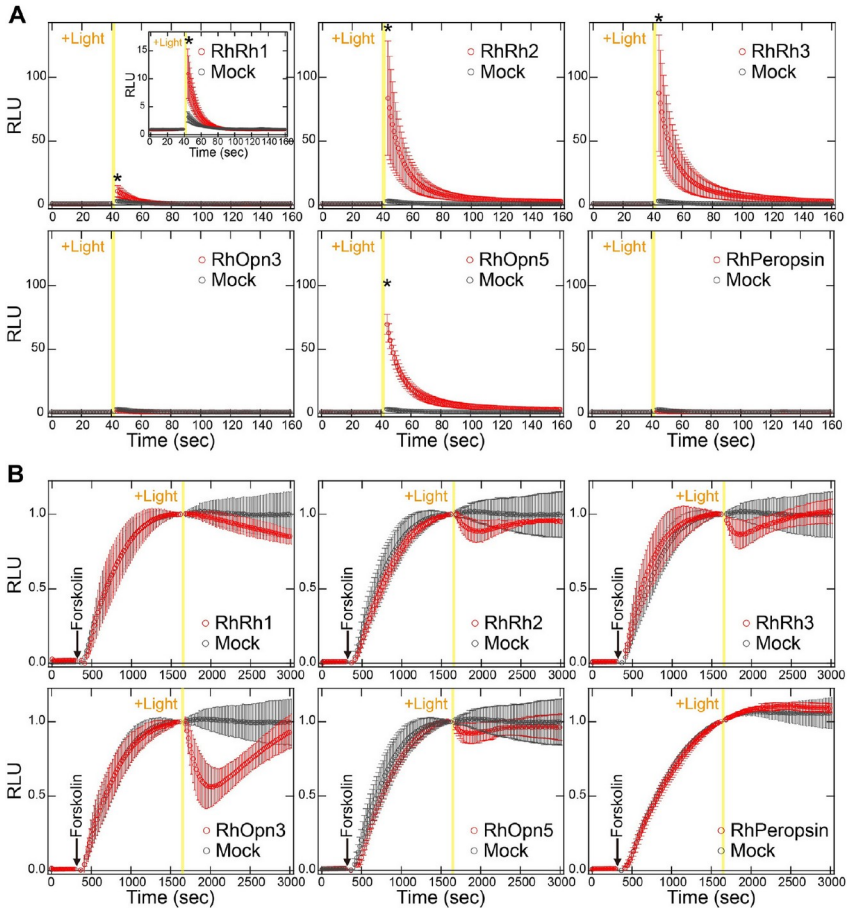
Nagata et al. (2025), Fig. 3. Opsin characterization and light-dependent responses.

A light-sensitive receptor engages cellular signaling

A **signaling pathway** is a sequence of cellular events linking receptor activation to a measurable response. Nagata and colleagues used cellular assays to determine whether illumination of expressed shrimp opsins altered downstream signaling.

The Opn5 receptor formed a blue-light-sensitive pigment and engaged inhibitory G-protein signaling. A **G protein** couples an activated receptor to intracellular effectors. Cells expressing this receptor showed a light-dependent decrease in cyclic AMP, a molecule that carries intracellular signals.

Cells without an introduced opsin provided a comparison for the light-dependent measurements. The researchers also measured intracellular calcium responses to examine another signaling route. These preparations connect pigment activation to biochemical changes within cells, rather than treating photon absorption as the endpoint of sensory reception.



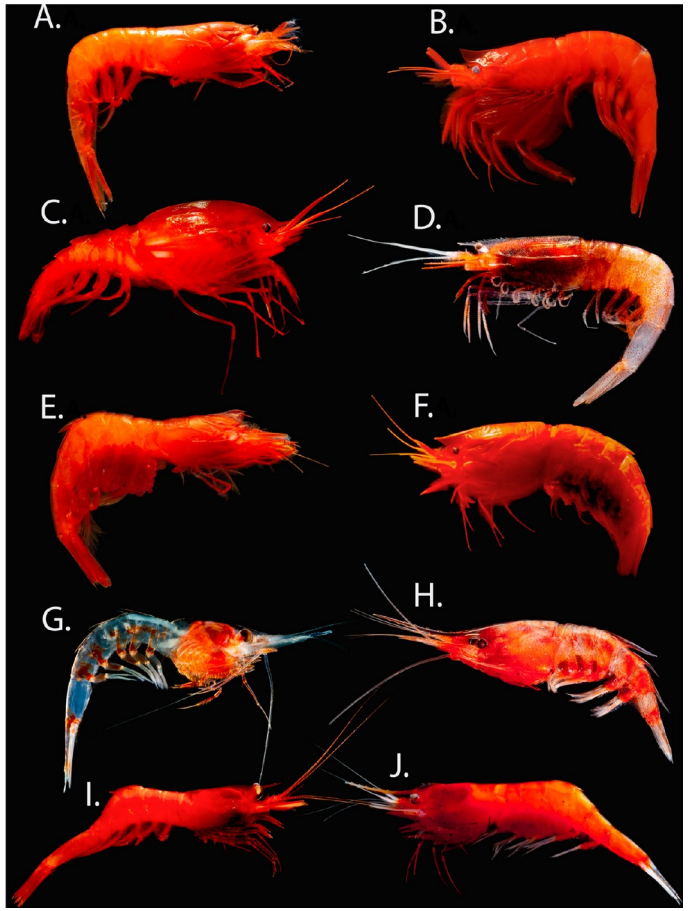
Nagata et al. (2025), Fig. 2. Light-dependent calcium and cyclic AMP responses.

Migration exposes shrimp to changing light environments

Diel vertical migration is movement through the water column associated with the daily light cycle. DeLeo and Bracken-Grissom compared opsin repertoires among bioluminescent oplophoroid shrimp with different migration patterns and light-producing structures.

Animals reaching shallower water encounter a more variable light environment than animals remaining deeper. Their visual systems may receive both changing background illumination and brief light signals produced by nearby organisms.

The comparative study found higher opsin diversity among species migrating into shallower, more variable conditions. This connects environmental variation to molecular receptor diversity, providing a crustacean comparison with the expanded visual receptor repertoires described in some deep-sea fishes.



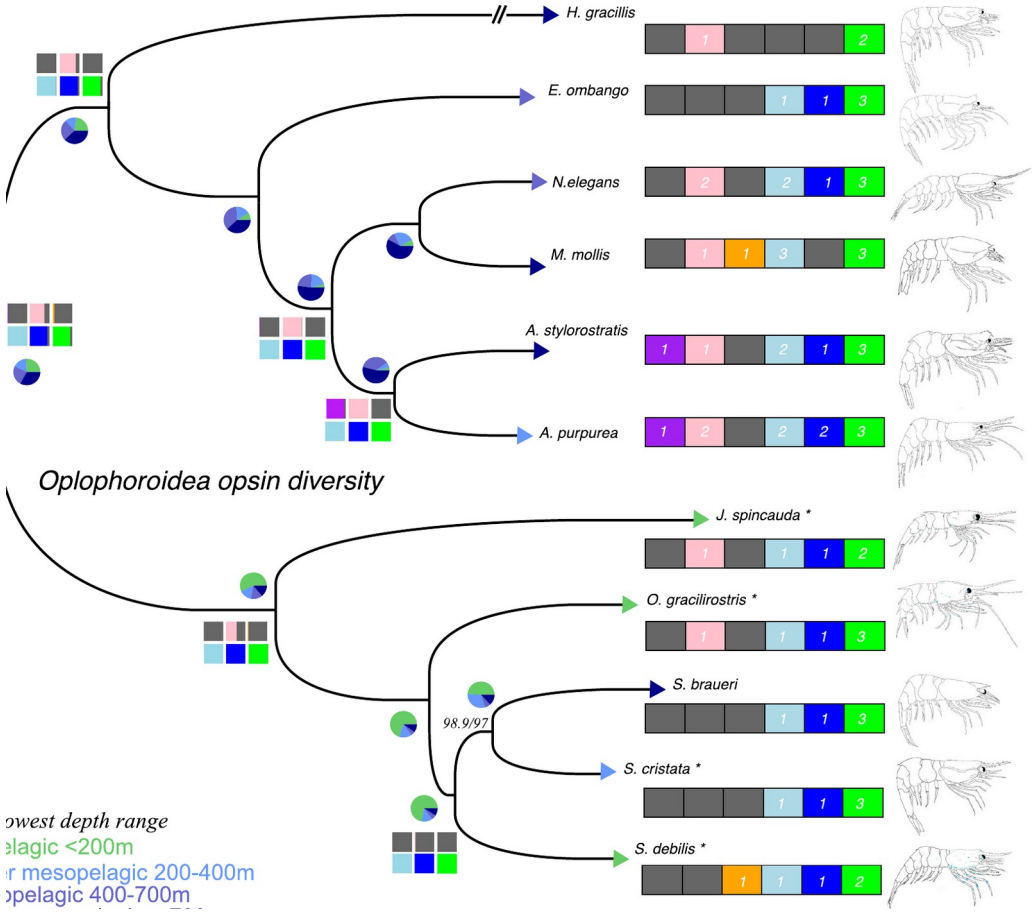
DeLeo & Bracken-Grissom (2025), Fig. 1. Opsin evolution and shrimp sensory ecology.

Light-producing shrimp also diversify light reception

Oplophoroid shrimp differ in their bioluminescent equipment, including species with specialized photophores. DeLeo and Bracken-Grissom examined relationships among this anatomical variation, migration ecology, and the evolution of light-sensitive opsin sequences.

Photophore-bearing lineages exhibited diversification and evidence of selection involving a putative middle-wavelength-sensitive opsin. **Selection** here concerns evolutionary changes in receptor sequence associated with lineage history and sensory ecology.

The authors propose that the additional receptor diversity may help discriminate environmental light from bioluminescent sources. This source-discrimination hypothesis links light emission and reception, while the measured comparative evidence concerns receptor sequences and their distribution among shrimp species.



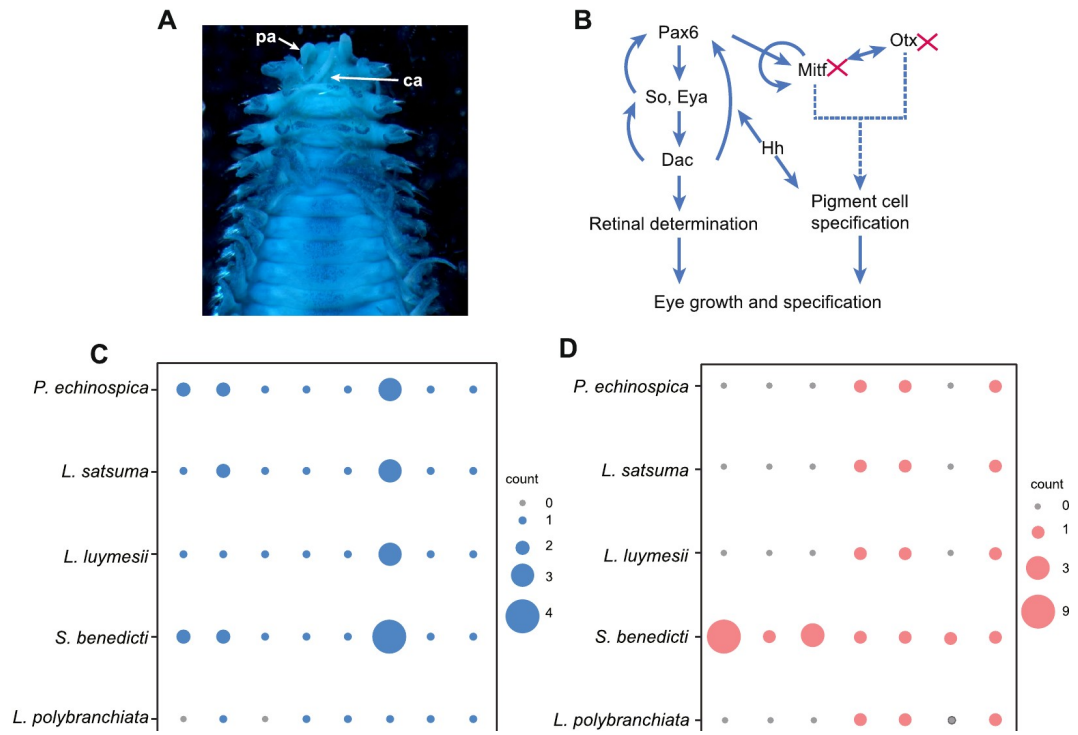
DeLeo & Bracken-Grissom (2025), Fig. 2. Opsin evolution and shrimp sensory ecology.

Cold-seep worms show reduced visual equipment

The annelid *Lindaspio polybranchiata* inhabits deep-sea cold seeps, places where reduced compounds emerge from the seafloor. Yan and colleagues examined its anatomy and genome to investigate the sensory equipment of an animal living without ordinary surface illumination.

Visual degeneration describes reduction of components normally associated with vision. The study connected changes in visual structures and gene repertoires to the worm's environment, providing a comparison with deep-sea animals that retain elaborate light-detection systems.

Darkness therefore does not produce one universal anatomical response. Some lineages expand visual receptor repertoires, while others reduce visual equipment and retain or expand different sensory capabilities that better match the available environmental information.



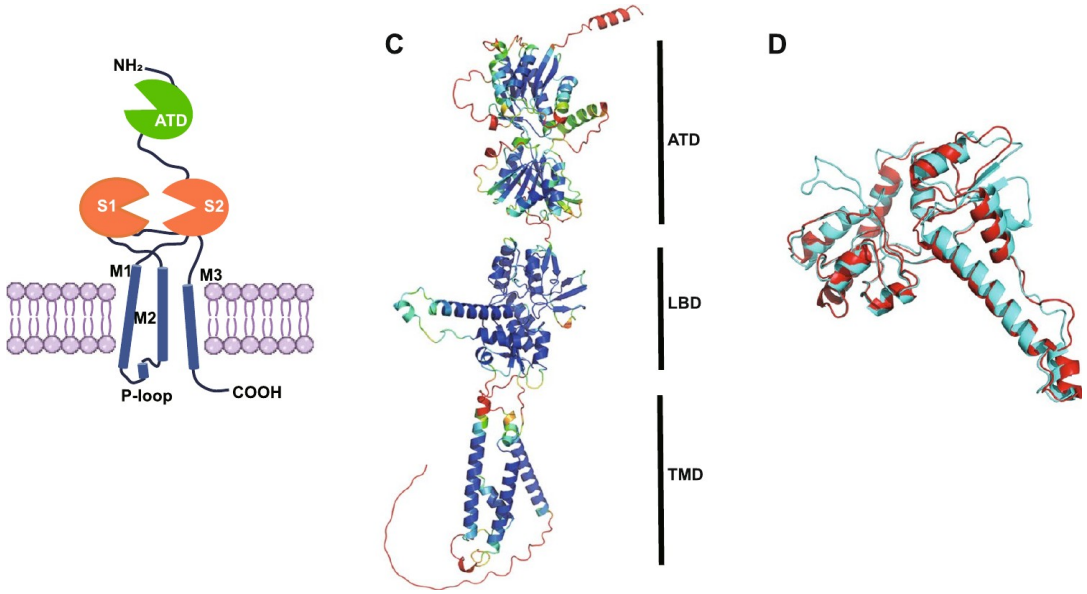
Yan et al. (2025), Fig. 2. Anterior anatomy and loss of vision-associated genes.

Chemoreceptor repertoires can expand in darkness

A **chemoreceptor** detects a chemical signal rather than a photon. Yan and colleagues identified expanded chemoreceptor gene repertoires in *Lindaspio polybranchiata*, alongside the reduction of visual equipment associated with its cold-seep environment.

Chemical information can identify local environmental conditions and potential resources where sunlight is unavailable. The genomic comparison connects the diversity of candidate chemical receptors to a habitat in which chemical gradients are a prominent source of information.

Receptor repertoire and sensory behavior are related at different levels of organization. The study supplies molecular and anatomical evidence for sensory specialization, complementing behavioral feeding studies that directly measure how other deep-sea animals orient toward chemical or contact cues.



Yan et al. (2025), Fig. 6. Chemosensory ionotropic receptor IR25a conservation.

Cirrate octopods alternate drifting and bottom activity

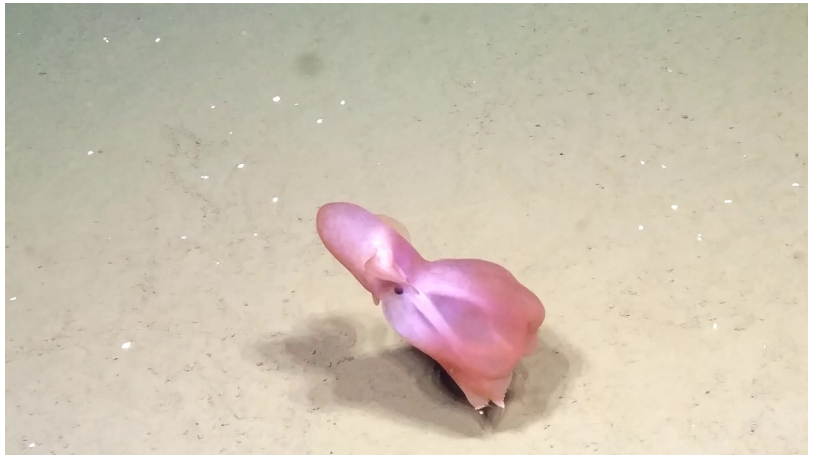
Cirroteuthis muelleri is a finned deep-sea octopod with a broad web between its arms. Golikov and colleagues documented individuals motionless in the water column and others active on the seafloor during Arctic deep-sea surveys.

Drifting individuals occurred hundreds to thousands of meters above the bottom. A spread web and reduced movement provide a behavioral state distinct from the repeated movements recorded during bottom-associated activity.

The authors propose a pelagic–benthic feeding migration, movement between the water column and the seafloor to obtain food. This interpretation links two observed behavioral states to the different resource and energetic conditions encountered away from and near the bottom.



Golikov et al. (2023), Fig. 2. Cirroteuthis muelleri posture and bottom interactions.



Golikov et al. (2023), Supplementary Video 02 (excerpt). Cirroteuthis muelleri near the Arctic seafloor at 3693 m.

A spread web envelopes a patch of seafloor

Bottom-associated *Cirroteuthis muelleri* performed a recurring sequence involving arm-web spreading, enveloping, and retreat. The web is the tissue connecting the arms, expanding the surface that can enclose a region beneath the animal.

Golikov and colleagues recorded sequences lasting 11–21 seconds. The ordered phases connect movement of a large anatomical structure to the location of potential prey on the bottom, giving the behavior a repeatable spatial and temporal organization.

The original video observations preserve the transition between open and enveloping postures. These transitions support the feeding interpretation because they repeatedly place the arm-web system around bottom material before the animal withdraws from the enclosed patch.



Golikov et al. (2023), Fig. 2. Cirrate-octopod posture and bottom interactions.



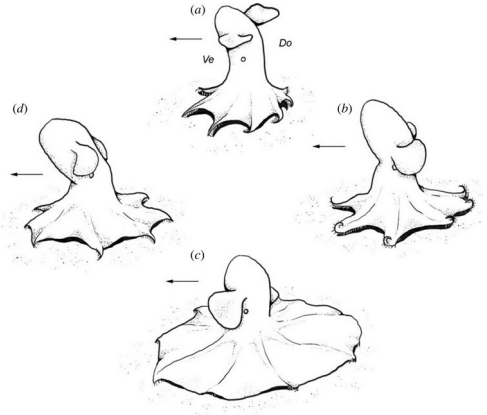
Golikov et al. (2023), Supplementary Video 01 (excerpt). Cirroteuthis muelleri spreads its web near the bottom at 3693 m.

Envelopment and retreat form a capture sequence

During the enveloping phase, the cirrate octopod brings its arm-web system around a section of the seafloor. Golikov and colleagues interpreted this phase as prey capture and recorded variable durations associated with the capture activity.

Reported capture phases lasted 5–49 seconds. The difference between the recurring overall movement sequence and the time spent enclosing a patch connects stereotyped motor organization to variation in local feeding circumstances.

Retreat returns the animal from the enclosed position to movement away from the patch. The sequence therefore couples encounter, enclosure, and withdrawal, contrasting with the filament-based collection of vampire squid and the pursuit-associated feeding of Humboldt squid.



Golikov et al. (2023), Fig. 3. Cirrate-octopod posture and bottom interactions.



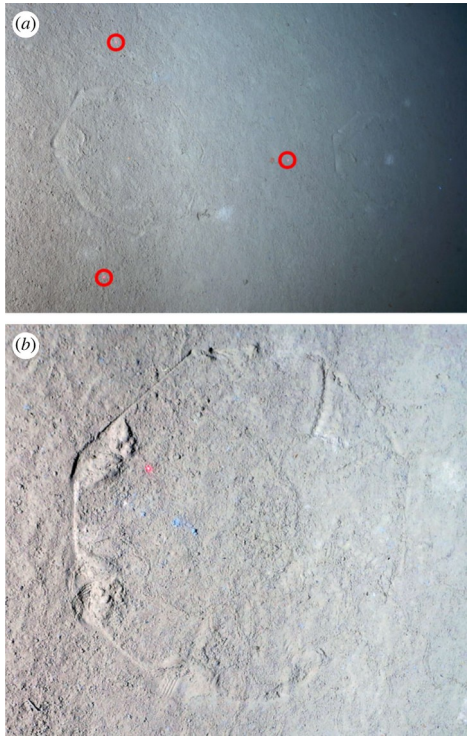
Golikov et al. (2023), Supplementary Video 03 (excerpt). Cirroteuthis muelleri contacts the bottom and retreats at 1506 m.

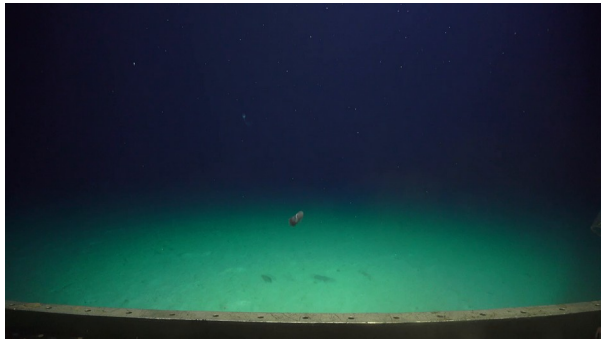
Seafloor traces record repeated bottom interactions

Golikov and colleagues recorded numerous traces associated with cirrate-octopod activity on the seafloor. A **feeding trace** is a physical mark linked to an animal's interaction with bottom material during the proposed feeding behavior.

Such traces connect a brief recorded encounter to a spatially broader pattern of activity. Repeated marks suggest that bottom interaction is an ecologically recurring behavior rather than a single unusual posture captured during one observation.

The combined evidence includes drifting animals, repeated bottom sequences, and associated traces. Together these observations support an interpretation in which a deep-sea octopod uses the water column for transport or resting and the seafloor as a feeding environment.





Golikov et al. (2023), Fig. 4. Seafloor traces of octopod bottom activity.

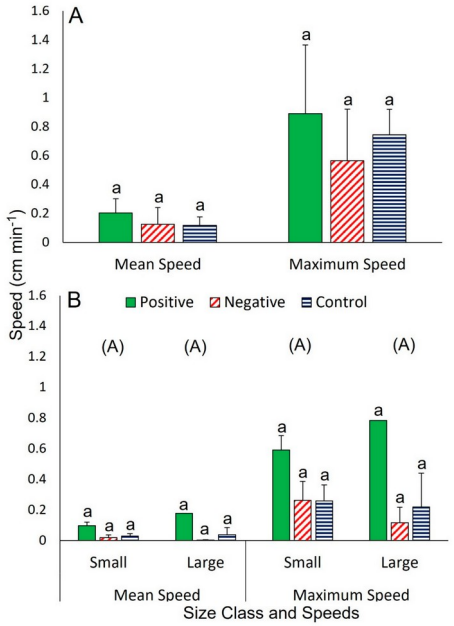
Golikov et al. (2023), Supplementary Video 04 (excerpt). Cirroteuthis muelleri moves at and above the bottom at 3615 m.

Food approach can be tested under dark conditions

Stuckless and colleagues studied deep-sea sea stars and a gastropod in controlled, darkened experiments. Food treatments and control conditions allowed comparison of directed approach with the movement animals made without the same food stimulus.

Chemosensory orientation is movement guided by chemical information associated with a stimulus. Approaching a food source in darkness provides behavioral evidence relevant to this sensory route, while contact with neighboring animals can modify the resulting path.

The experiments also varied the number and relative size of animals present. Food detection, locomotion, and access to food can therefore be examined separately from the assumption that the largest individual always secures the resource.





Stuckless et al. (2023), Fig. 1. Food approach, feeding and animal interactions.

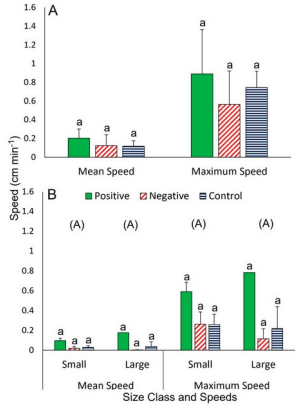
Stuckless et al. (2023), Supplementary Video S6. Henricia lisa travels toward food.

Contact changes access to a shared food source

Ceramaster granularis can change its route or position when another sea star occupies the path or food. Stuckless and colleagues recorded blocking, wedging, and shared access during experiments involving animals of different relative sizes.

Wedging is movement under another animal to gain access to the food beneath it. This tactic connects arm and body contact to a change in feeding position, rather than requiring displacement of the entire competitor.

The supplementary recording preserves interactions among individuals around the same food source. Their positions and movements connect encounter with a conspecific to access decisions, explaining why body size alone does not determine the outcome of competition in this preparation.



Stuckless et al. (2023), Fig. 1. Food approach, feeding and animal interactions.



Stuckless et al. (2023), Video S2. Blocking and shared feeding by Ceramaster granularis.



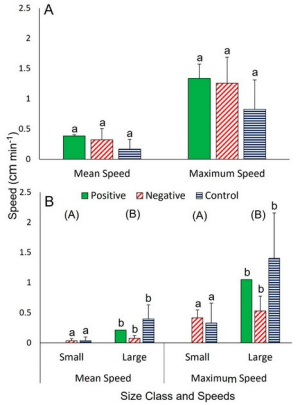
Stuckless et al. (2023), Supplementary Video S1. Small C. granularis wedges under a larger animal.

Some sea stars skirt around a neighbor

Hippasteria phrygiana often approached food along a straight path until another individual occupied the route. Stuckless and colleagues recorded animals moving around neighbors rather than consistently climbing across their bodies.

Skirting changes the path while preserving approach toward the resource. Brief contact between arm tips accompanies this interaction, linking local information from another animal to adjustment of locomotor direction.

The behavior contrasts with wedging in *Ceramaster granularis*. Coexisting deep-sea sea stars use different contact-related tactics around a shared resource, so a common food stimulus can produce species-specific movement sequences rather than a single universal competitive response.



Stuckless et al. (2023), Fig. 2. Food approach, feeding and animal interactions.



Stuckless et al. (2023), Video S4. Hippasteria phrygiana skirts a neighbor.



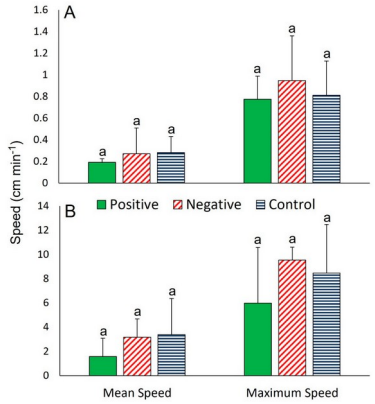
Stuckless et al. (2023), Supplementary Video S5. H. phrygiana pushes a smaller neighbor off food.

Arm-tip tapping samples food during handling

Henricia lisa displayed repeated arm-tip contact with food. Stuckless and colleagues described the tip being applied to and lifted from the food with limited displacement, distinguishing this activity from continuous locomotion toward the source.

This **tapping behavior** included successive contacts separated by roughly three minutes and movements of about one centimeter. The temporal pattern provides a repeatable way to characterize how the animal interacts with food after reaching it.

Contact behavior links acquisition of local information to handling rather than to distant approach. The distinction separates orientation toward a resource from the actions used to inspect or consume it, both of which contribute to successful feeding in darkness.



Stuckless et al. (2023), Fig. 3. Food approach, feeding and animal interactions.



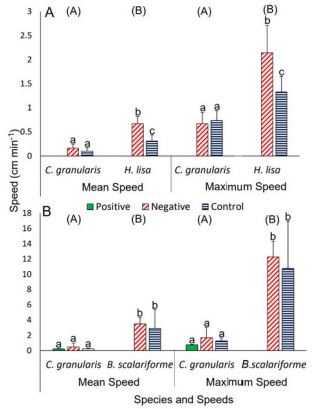
Stuckless et al. (2023), Video S7. Henricia lisa taps food with an arm tip.

A deep-sea gastropod follows food-directed paths

The gastropod *Buccinum scalariforme* approached food in the same experimental environment as the sea stars. Its movement uses a muscular foot rather than radial arms, providing a different anatomical route from sensory information to locomotion.

Stuckless and colleagues documented food-directed trajectories and interactions with other animals at the resource. The approach recording preserves the movement of the intact animal, complementing the trajectory and feeding measurements reported in the paper.

Comparing gastropod and sea-star behavior separates shared ecological demands from different motor organization. Both can locate and approach food under dark conditions, but their body structures constrain how they move around a neighbor and establish access to the resource.



Stuckless et al. (2023), Fig. 4. Food approach, feeding and animal interactions.



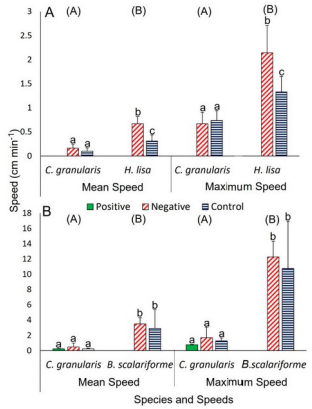
Stuckless et al. (2023), Video S8. Buccinum scalariforme approaches food.

Competition changes movement without a universal winner

The same resource can produce different outcomes depending on the species and individuals present. Stuckless and colleagues compared both conspecific encounters and interactions between deep-sea sea stars and the gastropod *Buccinum scalariforme*.

Some animals moved into space occupied by another feeder, while others changed position or retained access alongside a neighbor. **Interspecific interaction** means an encounter between species, and its motor sequence can differ from an encounter within one species.

Larger or faster animals did not invariably exclude smaller or slower ones. Access depends on food approach, body contact, handling tactics, and the arrangement of neighbors, connecting sensory-guided movement to a resource outcome rather than reducing behavior to body size.



Stuckless et al. (2023), Fig. 4. Food approach, feeding and animal interactions.



Stuckless et al. (2023), Supplementary Video S3. Large C. granularis pushes a smaller animal off food.



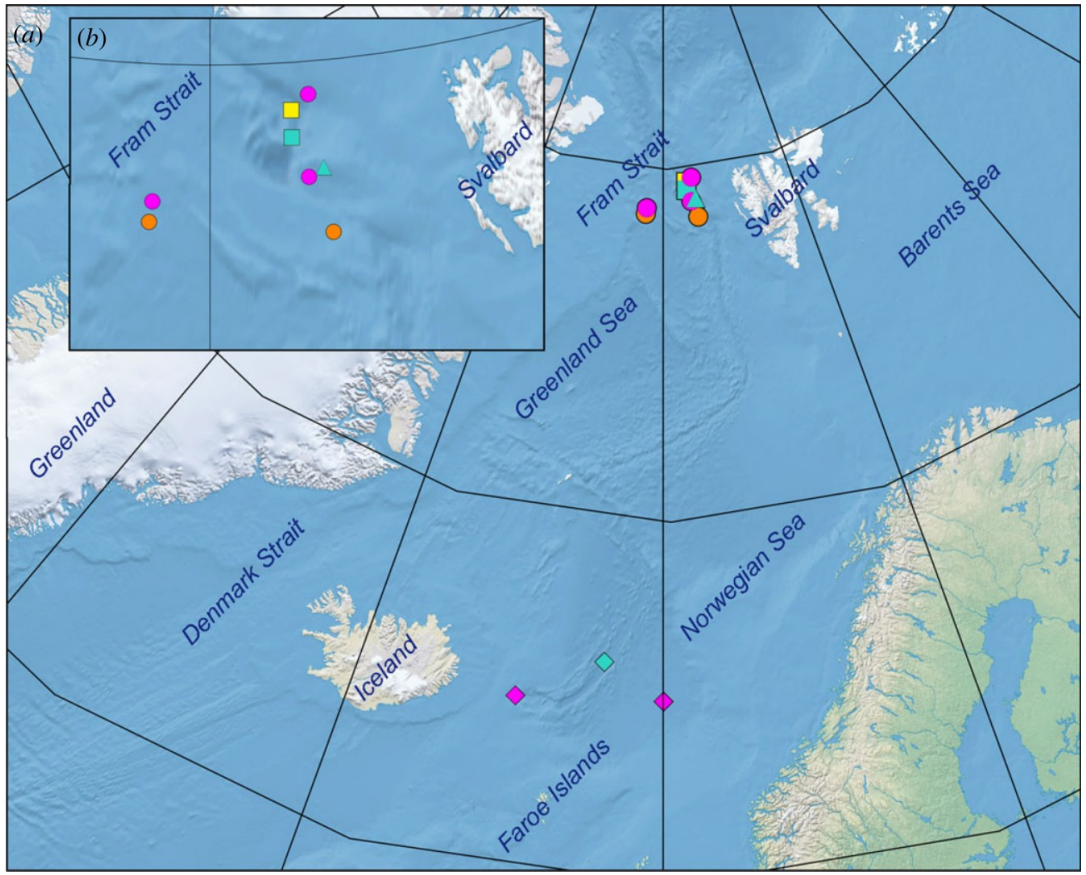
Stuckless et al. (2023), Supplementary Video S9. C. granularis wedges under H. lisa.

Sensory mechanisms require evidence at several levels

Deep-sea sensory adaptations can be examined through anatomy, receptor biochemistry, gene expression, and intact-animal behavior. Each preparation measures a different connection between environmental information and the actions available to an animal.

Pigment spectra connect molecules to wavelengths. Eye orientation connects sensory anatomy to the direction of light. Filament collection, food approach, and octopod envelopment connect body structures to resource acquisition under natural or experimentally controlled conditions.

Comparisons across fishes, cephalopods, crustaceans, annelids, echinoderms, and gastropods reveal multiple solutions to darkness and limited resources. Receptor diversification, directional sampling, chemical orientation, and contact-guided handling alter different steps between an environmental event and successful behavior.



Golikov et al. (2023), Fig. 1. Locations of Arctic octopod observations.

Key takeaways

Multiple sensory solutions. Deep-sea animals retain, diversify or reduce vision according to the signals available in their habitats.

Molecular tuning. Opsin sequence, developmental expression and accessory pigments alter the wavelengths available for detection.

Directional sampling. Cockeyed squids combine unequal eyes and maintained posture to sample silhouettes above and luminous sources below.

Production and reception. Photophores generate light; pigments and optical structures determine its visibility to another animal.

Nonvisual feeding. Filaments, arm-tip contact and chemical orientation connect darkness to species-specific collection and handling behavior.

Levels of evidence. Pigment measurements, anatomy and behavior establish different steps between environmental signals and animal actions.

Musilova et al. (2019); Thomas et al. (2017); Hoving & Robison (2012)